

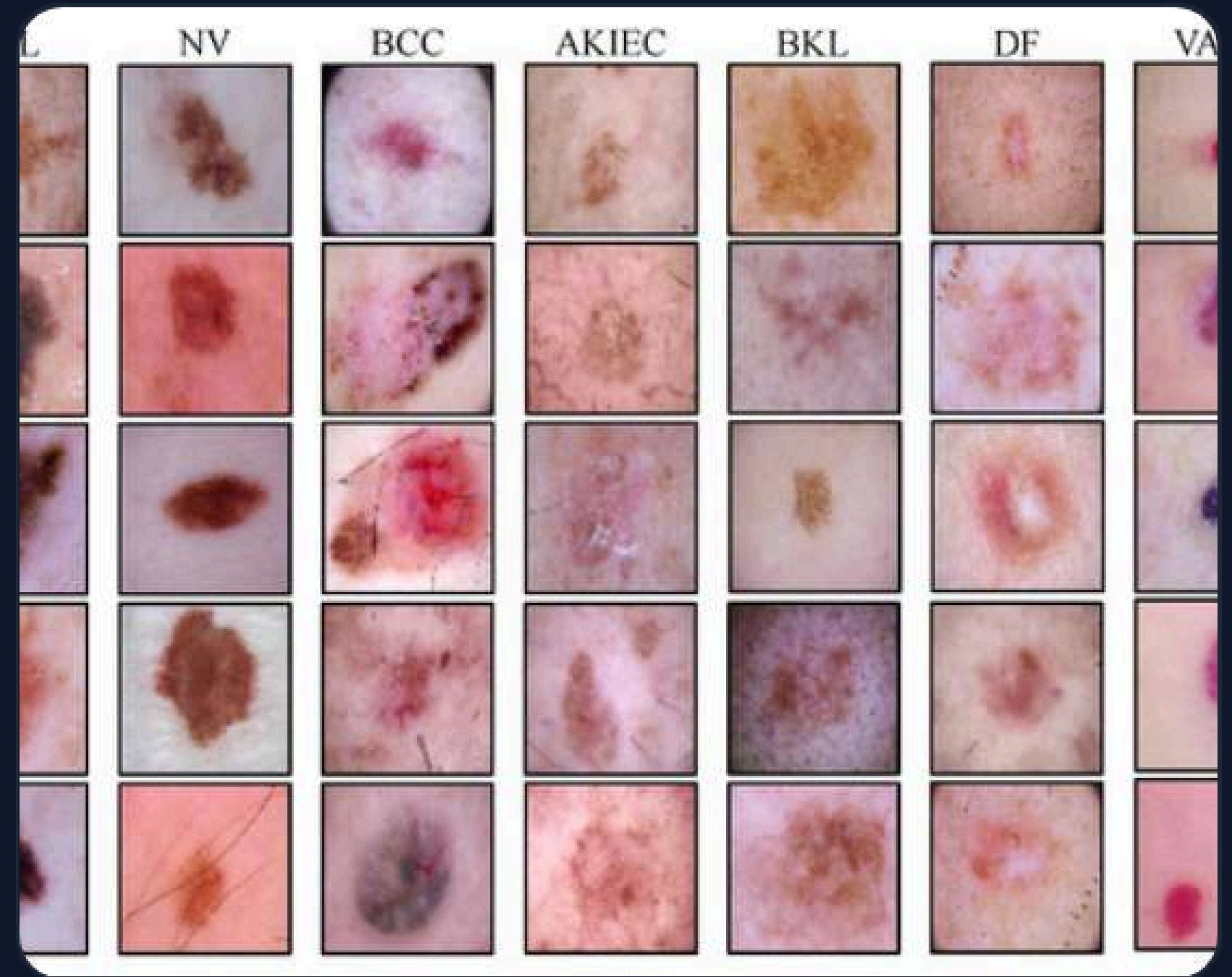
Skin Lesion Classification Using XAI

Explainable Artificial Intelligence Framework



Introduction & Motivation

- ✓ **Early Detection:** Critical for survival in melanoma cases; visual pattern recognition is key to clinical outcomes.
- ✓ **Dermoscopic Challenges:** Diagnosis is qualitative and morphological, often relying on complex visual sign structures.
- ✓ **Black-Box Limitation:** Deep learning models offer high performance but lack actionable clinical reasoning for medical experts.



| Problem Statement



Opaque Decision Making

CNNs utilize non-linear, high-dimensional representations of pixels that create a "semantic gap" between model logic and clinical terminology.



Clinical Risks

Non-interpretable predictions pose safety risks. Doctors cannot rely on an AI "oracle" without seeing the "why" behind high-stakes diagnoses.



XAI Necessity

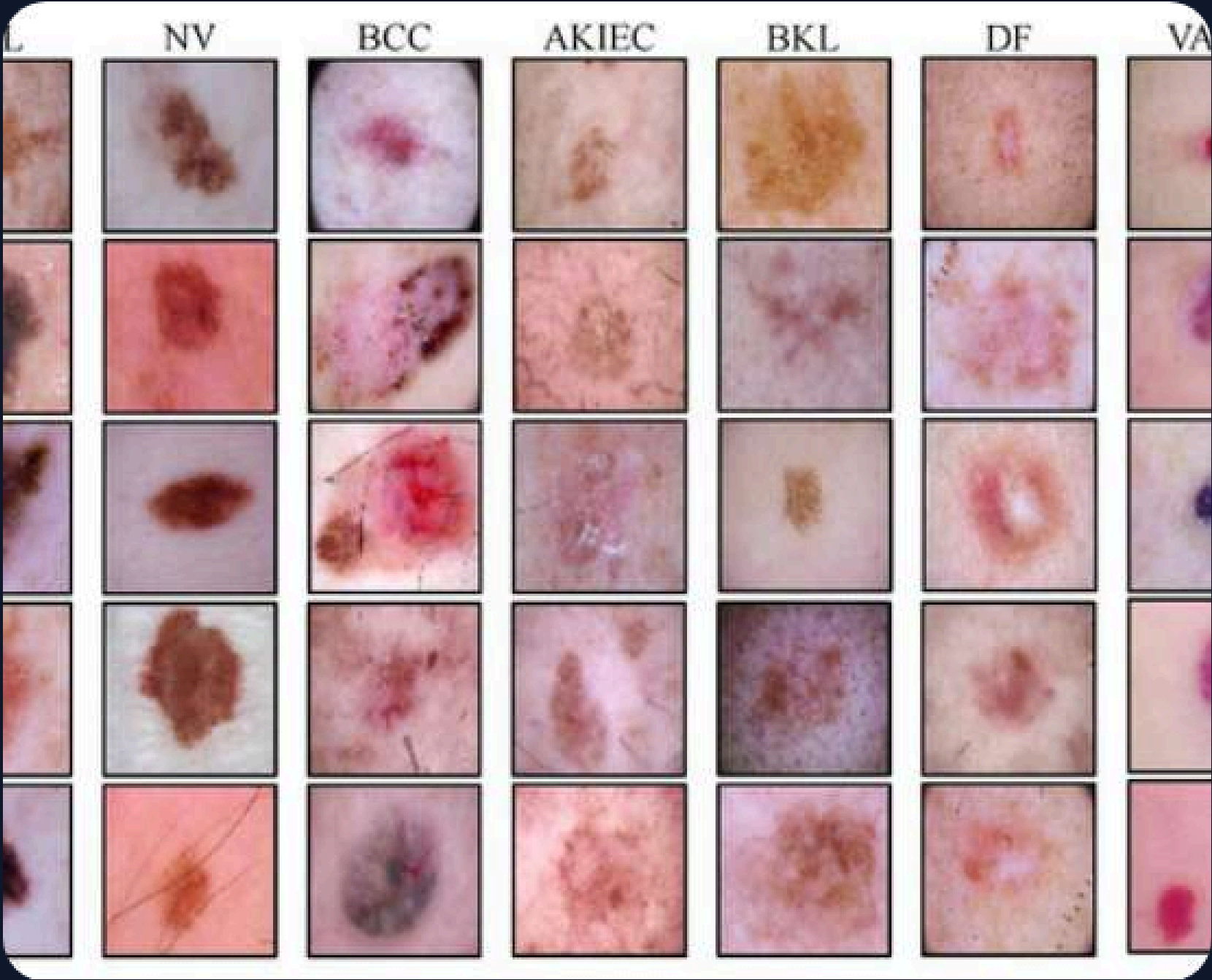
The need for Trust, Transparency, and Robustness as defined by FDA and EMA regulatory bodies for Software as a Medical Device (SaMD).

Dataset Overview

Source: Skin Lesion Segmentation (Hugging Face)

Total Population: 9,547 Dermoscopic Images

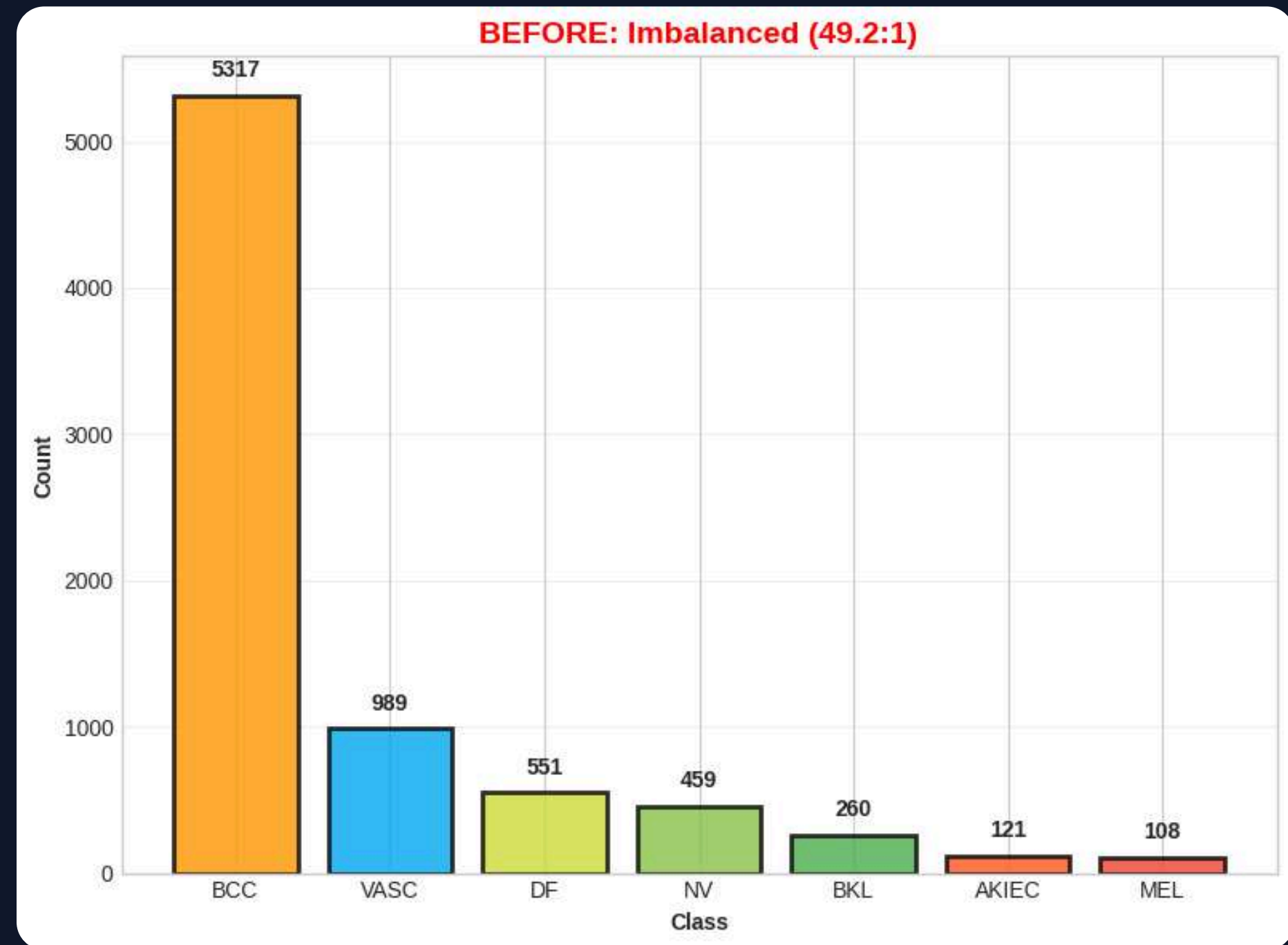
Split	Count	Purpose
Training	6,675	Learning Features
Validation	1,911	Hyperparameter Tuning
Testing	961	Final Performance Check



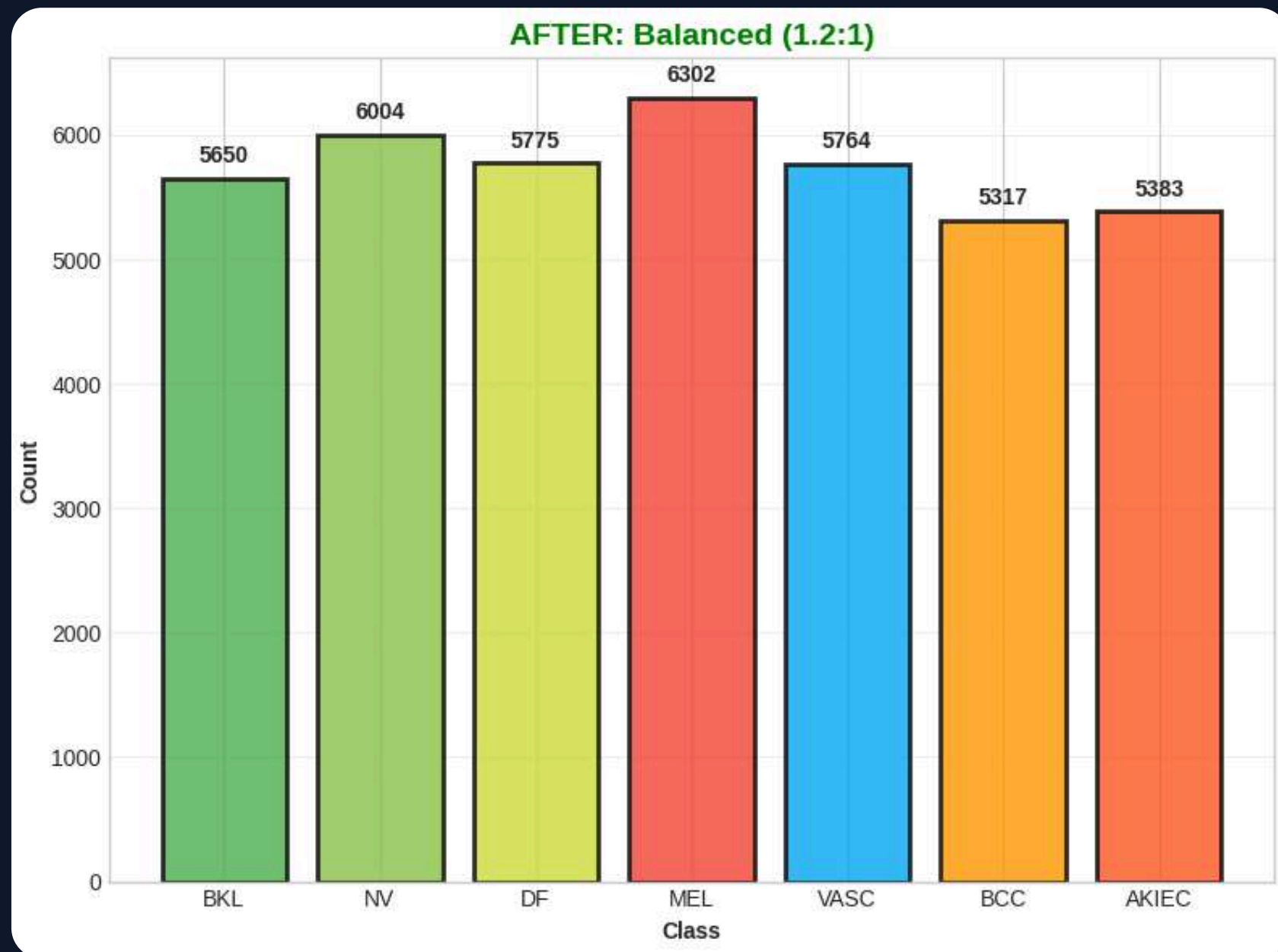
Class Distribution & Imbalance

The original dataset reflected real-world clinical rarity, creating extreme skew:

- ✓ **Imbalance Ratio:** Approximately 49.2 : 1
- ✓ **Clinical Risk:** High probability of false negatives for malignant minority classes (Melanoma).
- ✓ **Majority Class:** Benign Keratosis and Nevi dominated training.



Data Balancing Strategy



Fair Learning Implementation:

- ✓ **Downsampling:** Applied to the majority class to reduce dominance.
- ✓ **Synthetic Variation:** Minority classes augmented with 3 synthetic variations each.
- ✓ **Results:** Reduced imbalance ratio to 1.2 : 1.

Ensures model sensitivity to high-risk lesion categories instead of prior probability bias.

Performance Metrics Framework

Macro F1-Score

Evaluates model performance across minority and majority classes equally, exposing identifying failures.

AUC-ROC

Measures the model's separability and ranking capability between malignant and benign conditions.

Weighted Accuracy

Global correctness metric reflecting overall classification success on balanced validation sets.

"Macro metrics are essential to ensure the AI doesn't overlook life-threatening Melanoma while chasing global accuracy."

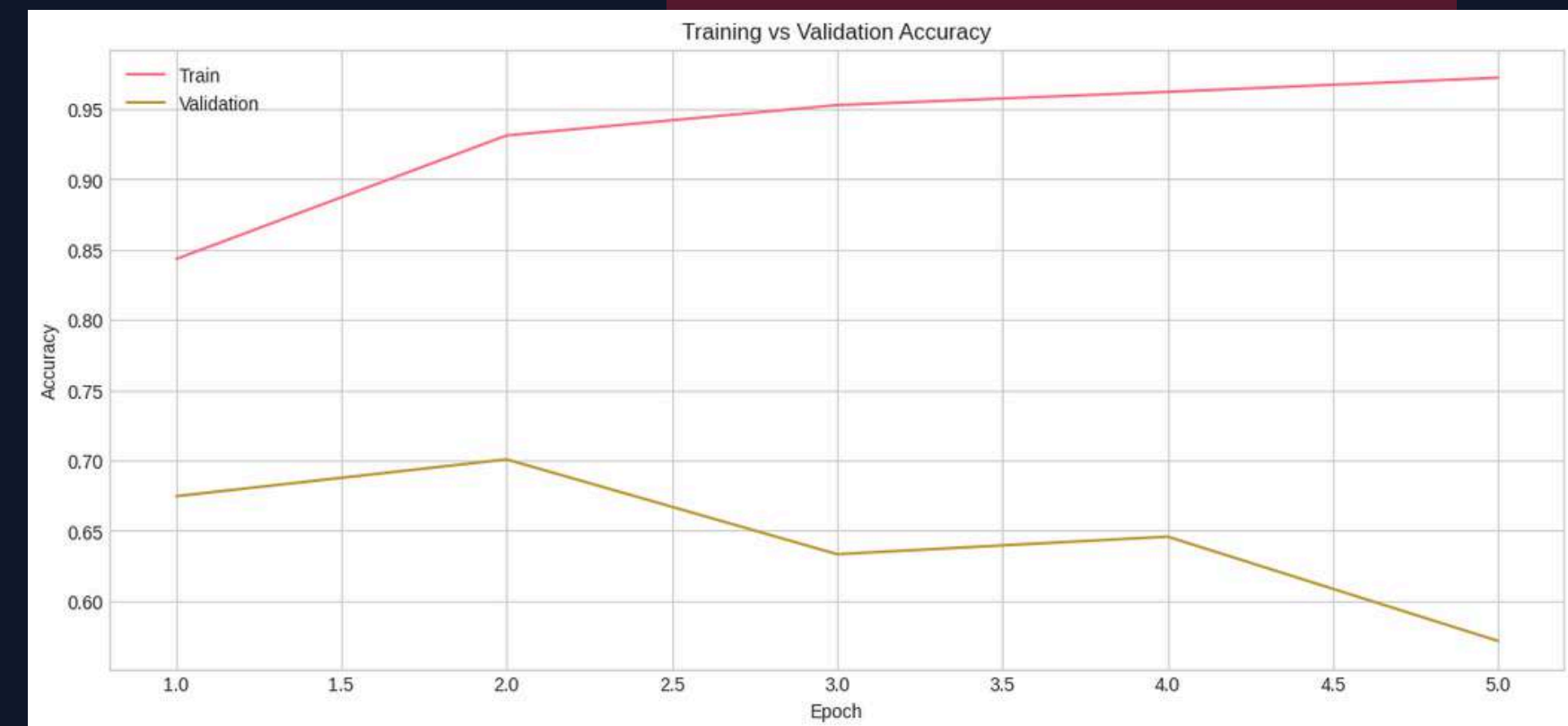
Quantitative Model Performance

57.1%
VAL ACCURACY

0.80
AUC SCORE

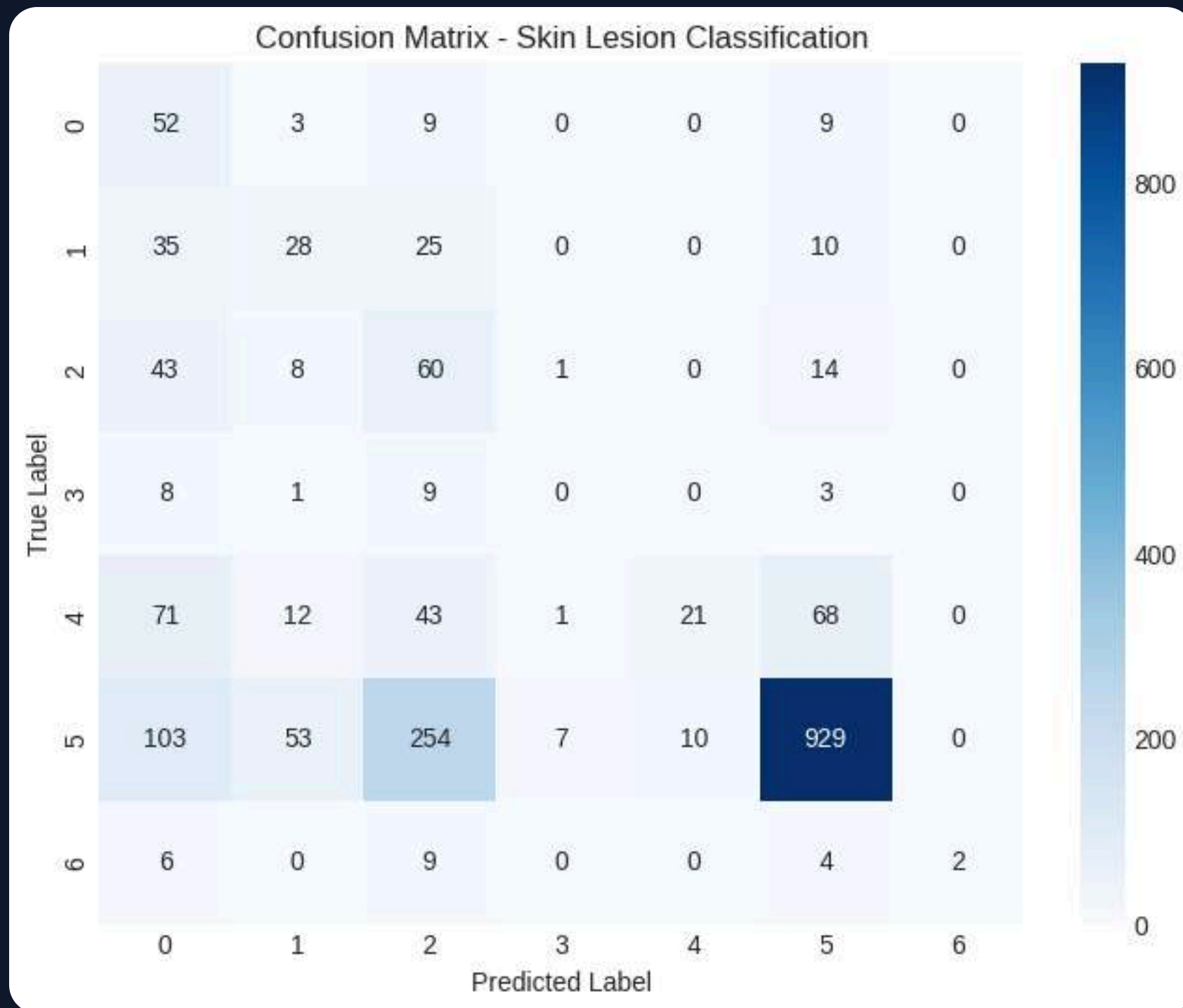
0.27
MACRO F1

2.10
RMSE



Evidence of **Overfitting**: High training accuracy (>95%) vs lower validation accuracy suggests the model memorized training samples.

Confusion Matrix Analysis



- ✓ **High Precision:** Class 5 (BCC) identified with 929 correct predictions.
- ✓ **Recall Gaps:** Class 2 (MEL) suffered from high false negatives.
- ✓ **Interpretation:** The model is learning ~~valid~~ features but struggles with decision thresholds for rare malignant cases.

| Why Explainable AI (XAI)?



Trust & Safety

Enables "sanity checks" to prevent False Negatives in deadly melanoma detection.



Decision Support

AI acts as a perceptual amplifier, drawing attention to subtle clinical markers.



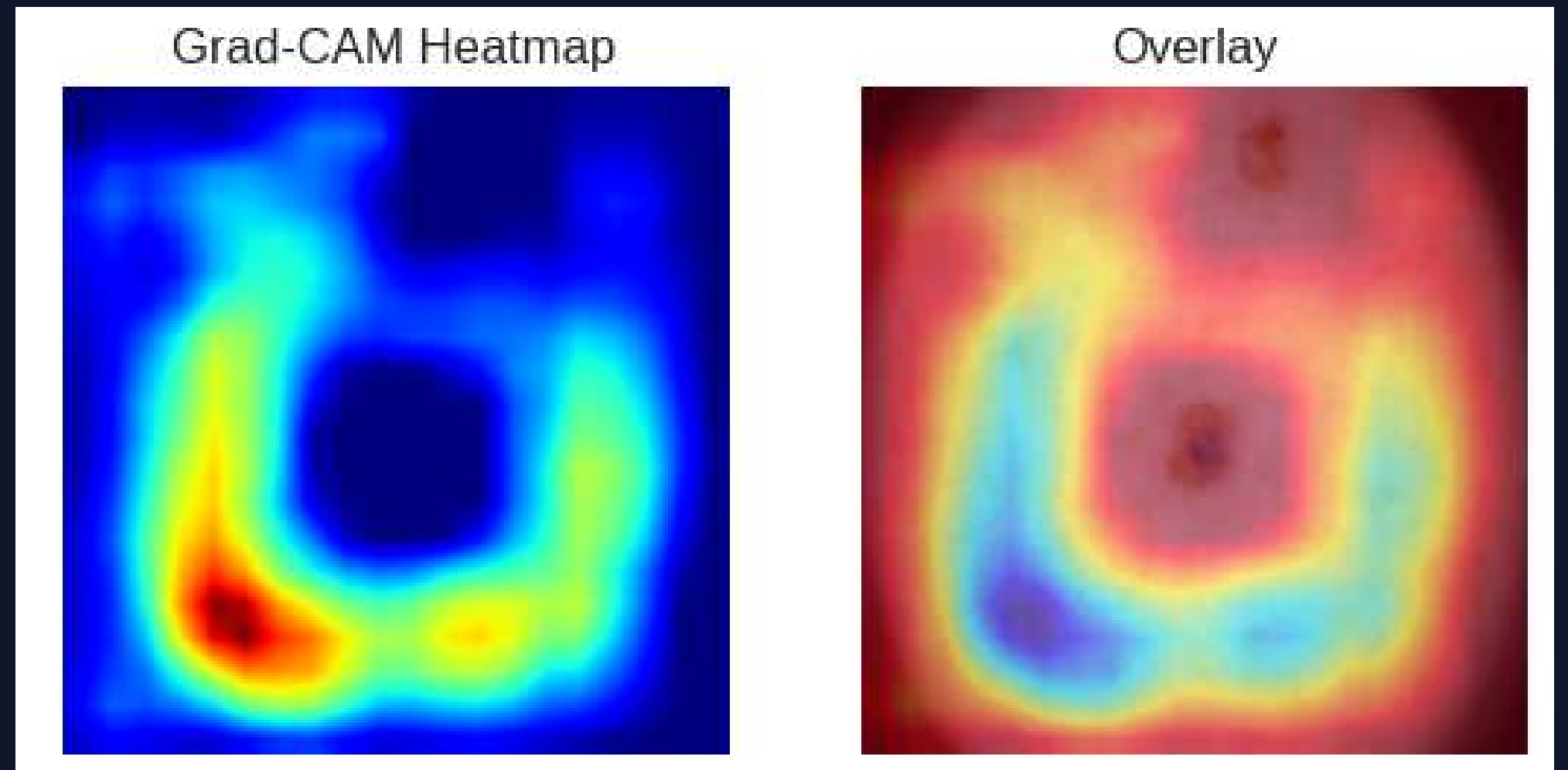
Regulatory Needs

Fulfills "Right to Explanation" requirements for high-risk medical AI systems.

Grad-CAM: Visual Feature Importance

Clinical Focus Validation:

- ✓ Uses gradient information to identify "hot" regions of model focus.
- ✓ **Successful Alignment:** Heatmap correctly localizes the pigmented lesion edge.
- ✓ **Evidence:** Confirms the model focuses on pathology, not background artifacts.



Counterfactual Explanation

Meaningful Perturbation

Causal Logic: "If I remove this central feature, does the diagnosis shift?"

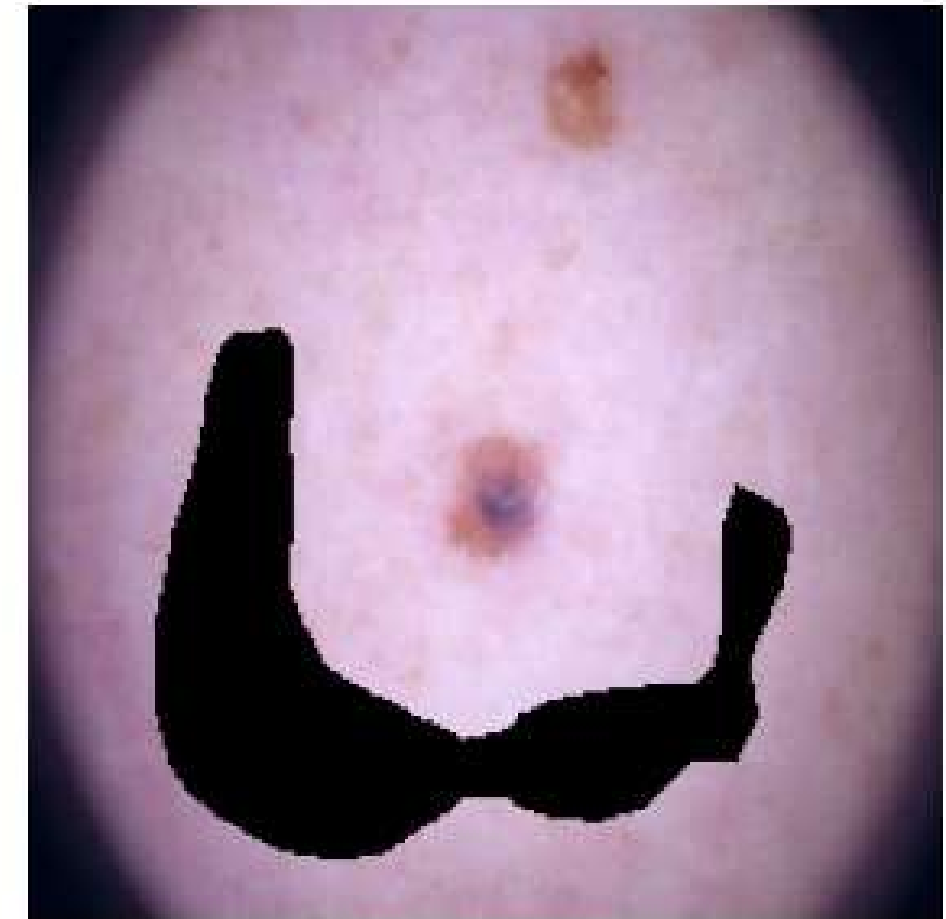
Key Finding: Removing the central white patch from Class 2 (DF) forced a 99% confident shift to Class 5 (BCC).

Reveals prediction fragility and high dependence on specific morphological structures.

Original (224x224)



Perturbed



Model Summary

Prediction

BKL

Confidence

85.86%

Accuracy

57.14%

AUC

0.797

F1 (Macro)

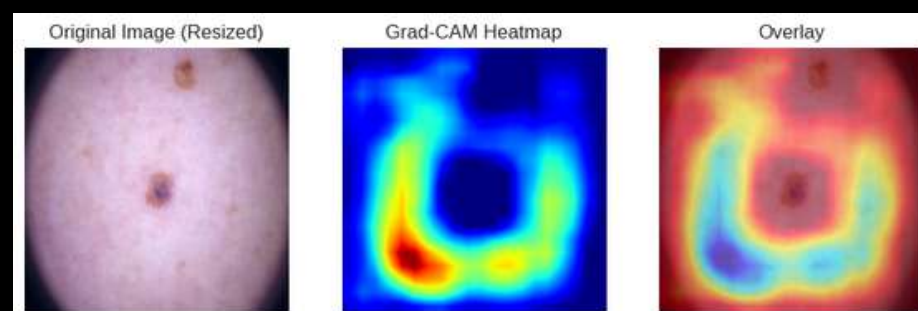
0.269

RMSE

2.107

CLINICIAN VIEW

GRAD-CAM EXPLANATION



CLINICAL INTERPRETATION

Grad-CAM highlights lesion boundaries and pigmentation patterns that strongly influenced the model's prediction. These regions align with dermatological diagnostic criteria.

✓ Supports clinical trust

PATIENT VIEW

COUNTERFACTUAL PERTURBATION



PATIENT-FRIENDLY EXPLANATION

Removing important lesion regions changes the model's confidence significantly, showing that predictions are based on visible medical features rather than random guessing.

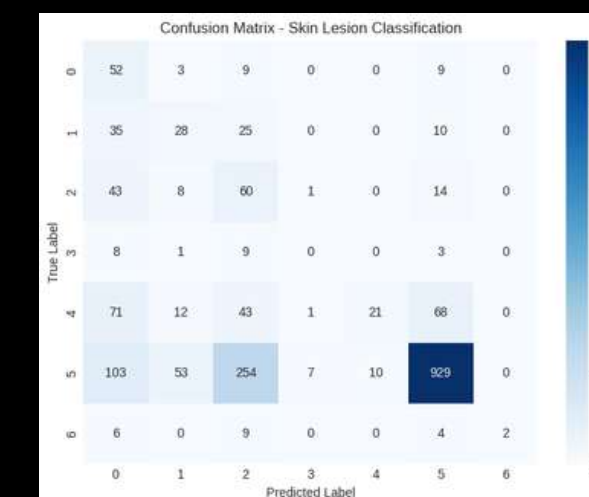
Before: 85.86%

After: 99.34%

✓ Improves understanding & reassurance

AUDITOR / REGULATOR VIEW

CONFUSION MATRIX



AUDIT INSIGHTS

- Strong performance on NV class
- Low recall on DF due to imbalance
- Bias identified and documented

⚠ Bias acknowledged & mitigated



Stakeholder Questions Addressed by XAI

Stakeholder	Key Question	XAI Technique Used	Answer Provided
Clinician	Why did the model predict this class?	Grad-CAM	Highlights lesion regions aligned with dermatology features
Patient	Can I trust this result?	Counterfactual	Confidence drops when lesion removed
Auditor	Is the model biased or unsafe?	Confusion Matrix + Metrics	Minority class risks identified

IMPACT ANALYSIS & DECISION SUPPORT



CLINICAL IMPACT

- REDUCES FALSE TRUST IN BLACK-BOX PREDICTIONS
- ENABLES VERIFICATION BEFORE DIAGNOSIS

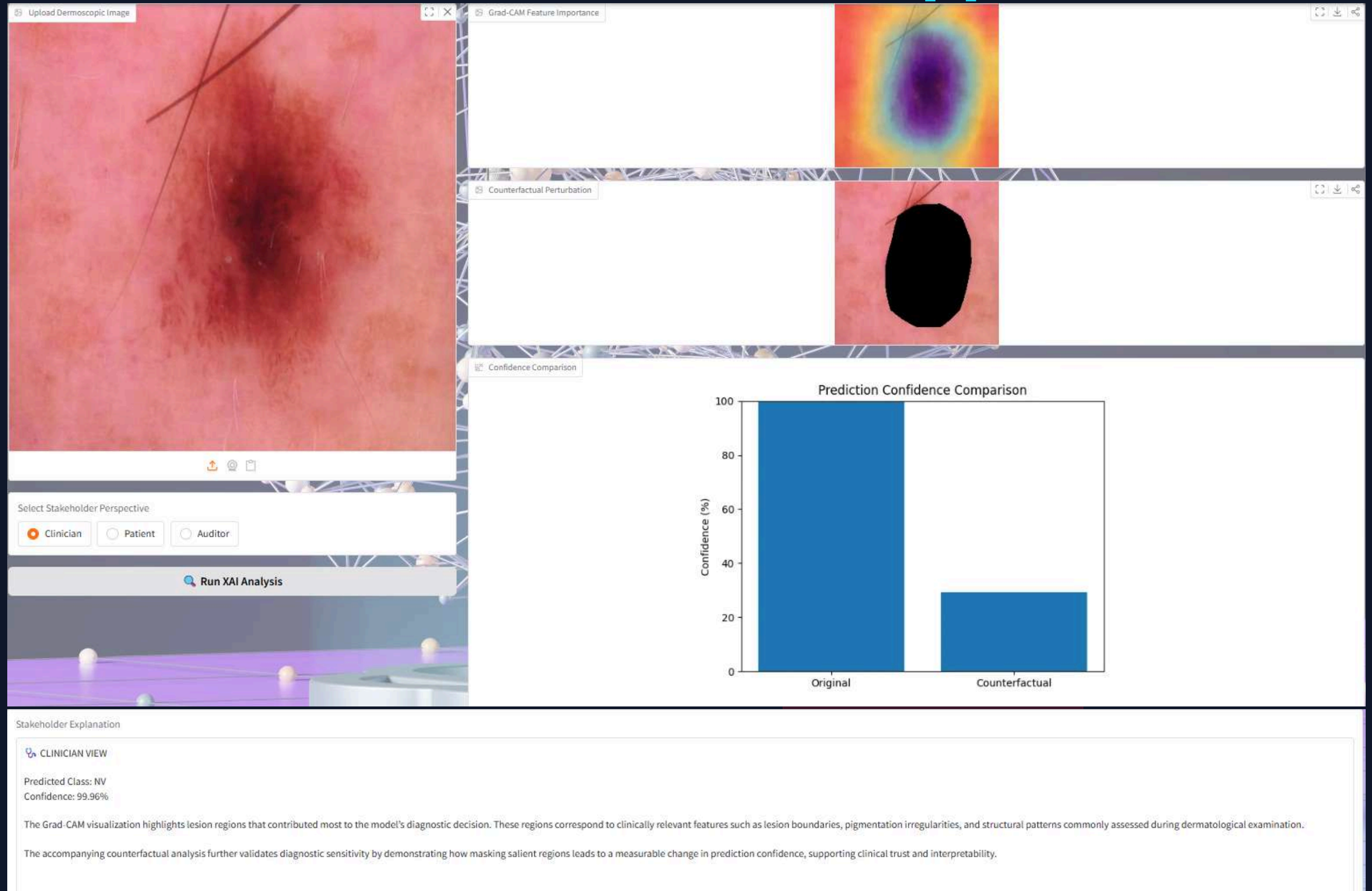
PATIENT IMPACT

- IMPROVES TRANSPARENCY AND UNDERSTANDING
- REDUCES ANXIETY VIA VISUAL EXPLANATIONS

REGULATORY IMPACT

- SUPPORTS "RIGHT TO EXPLANATION"
- DOCUMENTS BIAS AND FAILURE MODES

Interactive XAI Dashboard (Gradio App)



| Conclusion & Key Takeaways

Verification

Grad-CAM and Perturbation prove the model is learning meaningful visual features rather than background noise.

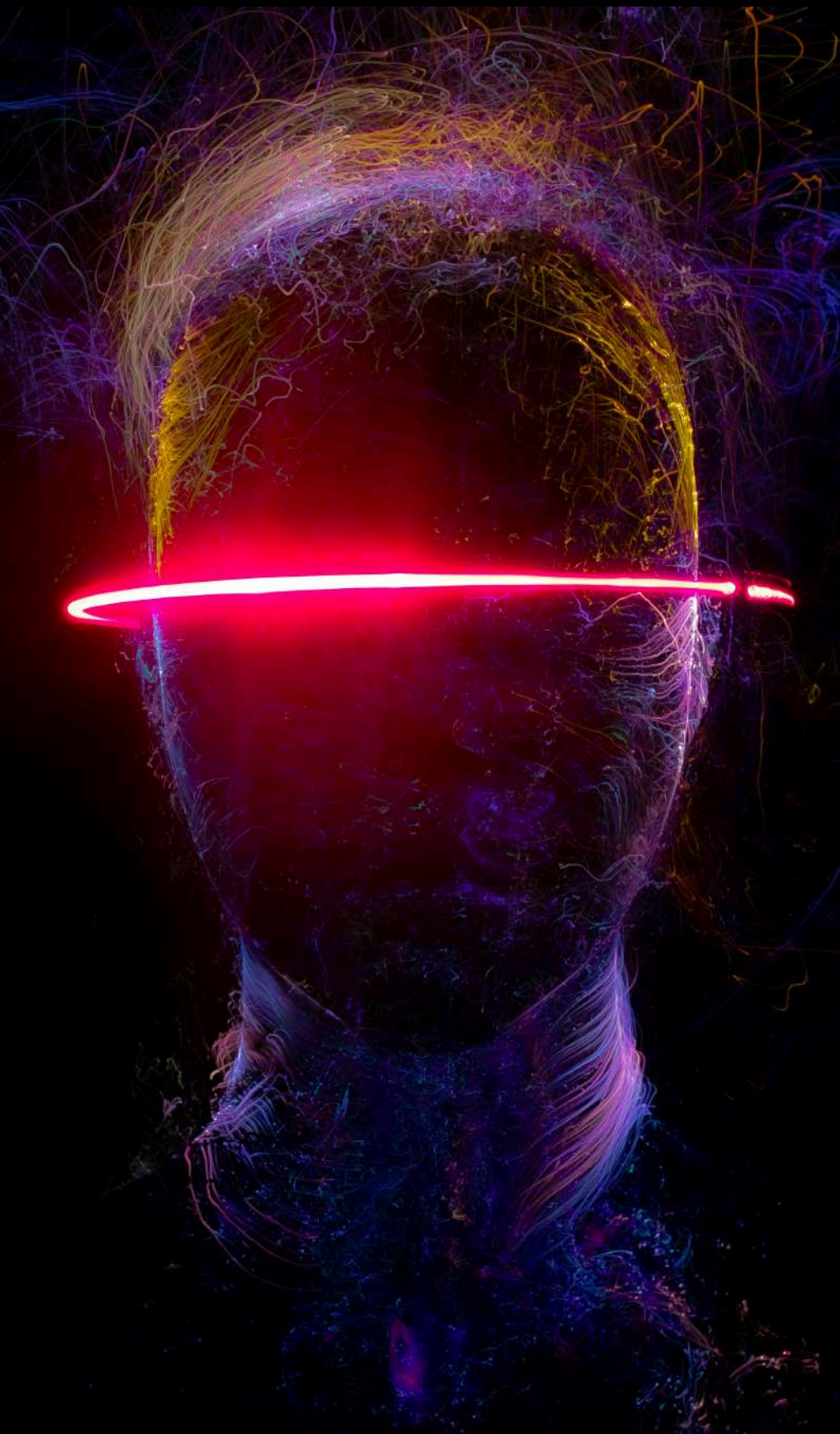
Trust

XAI tools shifted the focus from mere accuracy to trustworthiness, an essential step for clinical adoption.

Future Path

Overfitting and minority recall need improvement via advanced regularization and hybrid XAI strategies.

The Combined Clinician + XAI Accuracy > Either Alone



SALFORD & CO.

THANK YOU

CLOSING THOUGHTS AND ACTION

DASHBOARD LINK: SKIN LESION
CLASSIFICATION

